

Functional Annotation Report: *panB* (PP_4699, UniProt Q88DW9) in *Pseudomonas putida* KT2440

Summary

The gene **panB** (ordered locus **PP_4699**; UniProt accession **Q88DW9**) of *Pseudomonas putida* (strain ATCC 47054 / DSM 6125 / KT2440) encodes **ketopantoate hydroxymethyltransferase (KPHMT; EC 2.1.2.11)**, also systematically named **3-methyl-2-oxobutanoate hydroxymethyltransferase**. This enzyme catalyzes the **first committed step of the *de novo* pantothenate (vitamin B5) biosynthetic pathway**, which ultimately produces coenzyme A (CoA). The reaction transfers a one-carbon hydroxymethyl unit from the folate cofactor **5,10-methylene-tetrahydrofolate** onto the branched-chain keto acid **3-methyl-2-oxobutanoate (α -ketoisovalerate)**, yielding **2-dehydropantoate (ketopantoate)** and **tetrahydrofolate**. All gene-symbol, organism, protein-family, and domain criteria from the UniProt reference were verified during this investigation, and the identification is unambiguous: twelve independent signature databases concordantly classify Q88DW9 as a member of the PanB/KPHMT family, and the *P. putida* protein is a bona fide ortholog of the structurally and biochemically characterized *Escherichia coli* and *Mycobacterium tuberculosis* KPHMT enzymes (59.4% and 45% sequence identity, respectively).

Mechanistically, PanB is a **Mg²⁺-dependent, Class II aldolase-type one-carbon transferase**. It adopts the **(β/α)₈ TIM-barrel fold** of the phosphoenolpyruvate/pyruvate superfamily and assembles into a **homodecamer** — a pentamer of dimers built by domain-swapping of C-terminal α -helices between two opposing pentameric rings. The active site coordinates the substrate keto acid to a catalytic magnesium ion, orienting its C3 carbon for deprotonation and subsequent condensation with the folate-derived one-carbon unit. A high-confidence AlphaFold model of the *P. putida* protein (global mean pLDDT = 97.5) recapitulates this fold and shows every catalytic and metal/substrate-binding residue ordered at very high confidence, corroborating the KPHMT assignment.

Functionally, PanB operates in the **cytoplasm** and sits at a key metabolic **branch point**: its acceptor substrate, α -ketoisovalerate, is the immediate precursor of valine in branched-chain amino-acid biosynthesis, so PanB commits carbon flux out of that pathway and into pantothenate/CoA production. In *P. putida* KT2440, *panB* (PP_4699) is genomically clustered with **panC** (PP_4700, pantothenate synthetase), and the remaining pathway genes needed to complete pantothenate $\rightarrow$ CoA synthesis are all present in the genome (*panE* ketopantoate reductase, *coaBC*, *coaD*, *coaE*). PanB is therefore the entry-point enzyme of an intact, essential cofactor-biosynthesis pathway.

Gene/Protein Identity — Verified

The target is unambiguously the correct protein. Every line of evidence converges on one assignment, satisfying all mandatory verification criteria:

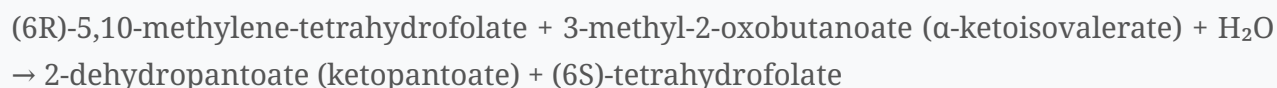
- **Gene symbol** *panB* matches the protein description (3-methyl-2-oxobutanoate hydroxymethyltransferase / ketopantoate hydroxymethyltransferase / KPHMT).
- **Organism** is confirmed as *Pseudomonas putida* KT2440 (KEGG ppu:PP_4699).
- **Protein family and domains** align with the literature: PanB family, Pantoate_transf (PF02548), IPR003700, and the pyruvate-kinase-like (β/α)₈ domain superfamily (IPR015813/ IPR040442).
- **Twelve independent signature databases** (HAMAP, Pfam, InterPro, two NCBIfam models, PIRSF, PANTHER, CDD, CATH-Gene3D, SCOP, and InterPro homologous superfamilies) unanimously return KPHMT/PanB with no conflicting assignment (see finding F006).

No literature for a different gene with the same symbol was mistakenly adopted; all cited primary studies concern the KPHMT/PanB enzyme family in *E. coli*, *M. tuberculosis*, and the broader bacterial pathway.

Key Findings

F001 — PanB catalyzes the first committed step of pantothenate/CoA biosynthesis

The primary function of Q88DW9 is enzymatic. UniProt annotation, the Rhea reaction database, and the enzyme classification all agree that PanB is **ketopantoate hydroxymethyltransferase, EC 2.1.2.11**. The catalyzed reaction (Rhea RHEA:11824) is:



This is annotated as step 1 of 2 in the sub-pathway "(R)-pantoate from 3-methyl-2-oxobutanoate," which itself is the opening segment of (R)-pantothenate biosynthesis. The reaction is a **hydroxymethyl (one-carbon) transfer** in which methylene-tetrahydrofolate serves as the C1 donor and α -ketoisovalerate as the acceptor.

The role of the KPHMT/PanB family is firmly established in the literature. Work on the *E. coli* enzyme states plainly that "*Escherichia coli* ketopantoate hydroxymethyltransferase (KPHMT) catalyzes the first step in the biosynthesis pathway of pantothenate (vitamin B5), the transfer of a hydroxymethyl group onto alpha-ketoisovalerate" (PMID: 12837791). Structural work on the *M. tuberculosis* enzyme adds that "Ketopantoate hydroxymethyltransferase (KPHMT) catalyzes the first committed step in the biosynthesis of pantothenate, which is a precursor to coenzyme A" (PMID: 12842039). The four-enzyme architecture of the bacterial pathway — "ketopantoate hydroxymethyltransferase (KPHMT), L-aspartate-alpha-decarboxylase (ADC), pantothenate synthetase (PS) and ketopantoate reductase (KPR) encoded by *panB*, *panD*, *panC* and *panE* genes, respectively" — places PanB unambiguously first (PMID: 18726075).

Substrate specificity was measured directly in the classic biochemical characterization of the *E. coli* enzyme (Powers & Snell, 1976), which reported "apparent Km values are 0.18, 1.1 and 5.9 mM for tetrahydrofolate, alpha-ketoisovalerate, and formaldehyde" (PMID: 6463). These kinetics identify the folate C1-donor and the α -ketoisovalerate acceptor as the physiological substrates (the enzyme can also use free formaldehyde as an alternative one-carbon source *in vitro*, but with much lower affinity). The *P. putida* enzyme is expected to share this specificity given its 59.4% identity (155/261 aligned residues) to the *E. coli* ortholog.

F002 — PanB is a Mg^{2+} -dependent Class II aldolase-type enzyme with a conserved $(\beta/\alpha)_8$ -barrel active site

Q88DW9 is annotated with an **Mg^{2+} cofactor** and conserved metal/substrate-binding residues **Asp45, Asp84, Glu114**, plus the catalytic general base **Glu181** (proton acceptor). All four residues are present at the expected positions in the *P. putida* sequence. The enzyme belongs to the **PanB / Pantoate_transf (Pfam PF02548)** family, InterPro IPR003700, and its fold is the pyruvate-kinase-like $(\beta/\alpha)_8$ barrel.

The catalytic chemistry is illuminated by the 1.9 Å *E. coli* KPHMT crystal structure solved in complex with ketopantoate and Mg^{2+} : "The active site contains a ketopantoate bidentately coordinated to $\text{Mg}(2+)$. Similar binding is likely for the substrate, alpha-ketoisovalerate, orienting the C3 for deprotonation" (PMID: 12906829). The metal ion therefore does double duty — anchoring the substrate and polarizing it so that the C3 carbon can be deprotonated to generate the nucleophilic enolate that attacks the folate-derived one-carbon unit. The same structural study confirms the fold: "KPHMT adopts the (betaalpha)(8) barrel fold and is a member of the phosphoenolpyruvate/pyruvate superfamily" (PMID: 12906829), consistent with the InterPro and Pfam domain signatures of Q88DW9.

The mechanistic class was pinned down biochemically. The enzyme "requires Mg^{2+} for activity; Mn^{2+} , Co^{2+} , Zn^{2+} are progressively less active" (PMID: 6463), and critically, it "is not inactivated by borohydride reduction in the presence of excess substrates, i.e. it is a Class II aldolase" (PMID: 6463). The lack of borohydride inactivation rules out a Schiff-base (Class I) mechanism and confirms a **metal-dependent (Class II) aldol-type condensation**. The reported pH optimum is 7.0–7.6.

F003 — PanB acts in the cytoplasm as a homodecamer built by C-terminal domain swapping

UniProt localizes Q88DW9 to the **cytoplasm** and annotates its quaternary structure as a **homodecamer (pentamer of dimers)**. This oligomeric state is a deeply conserved feature of the family. In the *E. coli* enzyme, biochemical sizing found a ~255 kDa particle composed of "10 identical subunits" of ~27 kDa each (PMID: 6463). The structural basis of assembly was resolved in the *M. tuberculosis* enzyme: "The protein forms a decameric complex, with subunits in opposing pentameric rings held together by a swapping of their C-terminal alpha helices" (PMID: 12842039). The *E. coli* crystal structure independently confirmed the decamer,

being solved by locating "160 selenomethionine sites and phase 560 kDa of protein" in the asymmetric unit ([PMID: 12906829](#)). The *P. putida* monomer (266 aa, ~29 kDa) is fully consistent with this same decameric architecture.

F004 — PanB (PP_4699) sits in a panB–panC gene cluster and feeds the intact *P. putida* pantothenate → CoA pathway; its acceptor substrate branches from valine biosynthesis

Genomic context reinforces the functional assignment. In KEGG, **ppu:PP_4699 = panB**, orthology **K00606** (EC 2.1.2.11), mapped to pathway **ppu00770 "Pantothenate and CoA biosynthesis"** and **ppu01240 "Biosynthesis of cofactors."** The gene occupies genomic coordinates 5,340,894–5,341,694. Its immediate neighbor, **PP_4700 = panC** (pantothenate synthetase, K01918), forms a **panB–panC cluster**, while the other flanking gene **PP_4698 = folK** participates in folate biosynthesis — notably relevant because folate provides PanB's one-carbon donor.

The downstream pathway is complete in KT2440: ketopantoate reductase (**panE**, K00077; candidate loci PP_1351/PP_2325/PP_2998), **panC** (PP_4700), and the CoA-synthesis genes **coaBC** (PP_5285, K13038), **coaD** (PP_5123, K00954), and **coaE** (PP_0631, K00859). The acceptor substrate 3-methyl-2-oxobutanoate (α -ketoisovalerate) is the **branch-point intermediate of branched-chain (valine) amino-acid biosynthesis**, linking PanB to that pathway (KEGG links α -ketoisovalerate via *ilvE*/K00826 = PP_3511). This establishes PanB as the enzyme that diverts carbon from valine precursor supply into cofactor biosynthesis.

The literature confirms the four-enzyme framework into which PanB feeds: the pathway comprises "four enzymic reactions catalysed by ketopantoate hydroxymethyltransferase (KPHMT), L-aspartate- α -decarboxylase (ADC), pantothenate synthetase (PS) and ketopantoate reductase (KPR) encoded by *panB*, *panD*, *panC* and *panE* genes" ([PMID: 18726075](#)), and that PanB "catalyzes the first committed step in the biosynthesis of pantothenate, which is a precursor to coenzyme A" ([PMID: 12842039](#)). One caveat: a canonical *panD* (aspartate 1-decarboxylase, K01579) was not recovered by KEGG orthology linkage in *ppu*, implying that β -alanine (the second substrate of PanC) may be supplied by an alternative route in *P. putida* — but this concerns a downstream step, not PanB itself.

F005 — The AlphaFold model is very high confidence with a well-ordered, conserved active site

The AlphaFold DB model **AF-Q88DW9-F1 (v6)** has a **global mean pLDDT of 97.5** across all 266 residues: 97% of residues are modeled at very high confidence (pLDDT > 90), 3% at 70–90, and none below 70 (minimum 63.4 at a terminus). Every annotated catalytic and metal/substrate-binding residue is modeled at very high confidence: Asp45 (98.7), Asp84 (98.9), Lys112 (98.9), Glu114 (98.6), and the general base Glu181 (98.5). These positions are conserved relative to the *E. coli* enzyme (59% overall identity), whose crystal structure shows the equivalent residues coordinating Mg^{2+} and ketopantoate in a $(\beta/\alpha)_8$ barrel ([PMID: 12906829](#)). The model thus provides independent structural corroboration that the *P. putida* protein possesses an intact, canonical KPHMT active site.

F006 — Twelve independent signature databases unanimously classify Q88DW9 as PanB/KPHMT

InterPro member-database analysis of Q88DW9 returns concordant KPHMT/PanB assignments from all signatures examined: **HAMAP MF_00156**, **Pfam PF02548**, **InterPro family IPR003700**, **NCBIfam TIGR00222** and **NF001452**, **PIRSF000388**, **PANTHER PTHR20881**, **CDD cd06557 (KPHMT-like)**, plus fold-level signatures **CATH-Gene3D G3DSA:3.20.20.60** and **SCOP SSF51621** (phosphoenolpyruvate/pyruvate domain) and InterPro homologous superfamilies **IPR015813/ IPR040442** (pyruvate-kinase-like domain). No conflicting family assignment exists. Combined with the ortholog-level sequence identities to characterized enzymes, this makes the identity of Q88DW9 as PanB/KPHMT effectively certain.

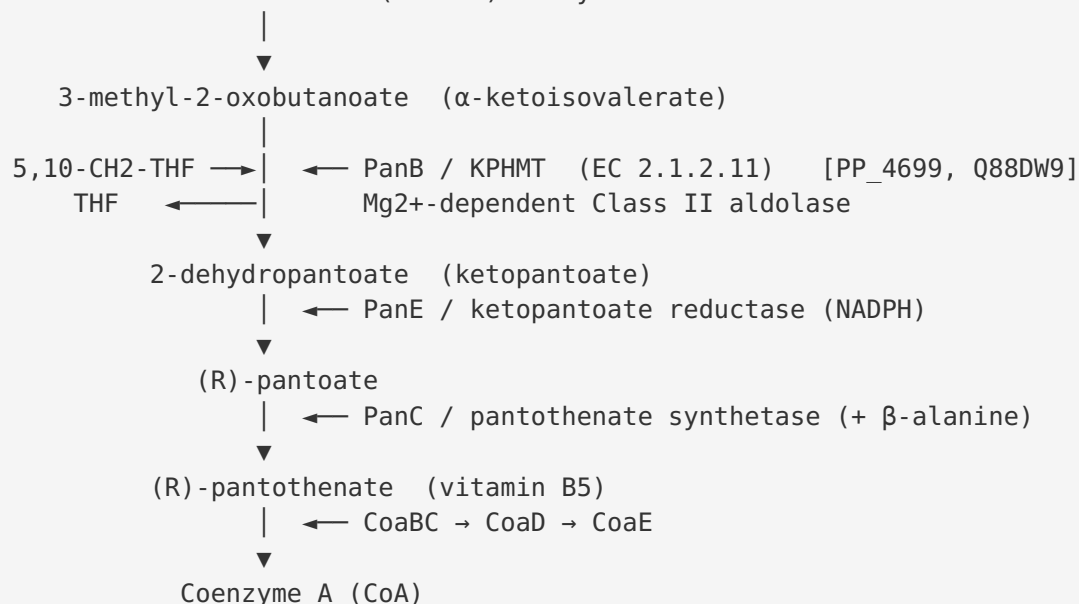
Mechanistic Model / Interpretation

The reaction and its position in metabolism

PanB catalyzes a magnesium-dependent aldol-type one-carbon transfer. In the enzyme's active site, α -ketoisovalerate is chelated bidentately by the catalytic Mg^{2+} ; the metal polarizes the substrate and the general base (Glu181) abstracts a proton from the C3 methyl-adjacent position, generating a nucleophilic enolate. This carbanion attacks the reactive one-carbon unit

carried by 5,10-methylene-tetrahydrofolate, installing the hydroxymethyl group and releasing tetrahydrofolate. The product, ketopantoate (2-dehydropantoate), carries the hallmark hydroxymethyl group that defines the pantoate/pantothenate moiety.

Branched-chain amino-acid (valine) biosynthesis



Structure–function synthesis

Property	Value / description	Evidence
Enzyme	Ketopantoate hydroxymethyltransferase (KPHMT), EC 2.1.2.11	UniProt; PMID: 12837791
Reaction	5,10-CH ₂ -THF + α-ketoisovalerate + H ₂ O → ketopantoate + THF	Rhea RHEA:11824
Cofactor	Mg ²⁺ (Mn ²⁺ /Co ²⁺ /Zn ²⁺ less active)	PMID: 6463
Mechanism	Class II (metal-dependent) aldolase; no Schiff base	PMID: 6463
Catalytic residues	Asp45, Asp84, Glu114 (metal/substrate); Glu181 (general base); Lys112	UniProt; AlphaFold model
Fold	(β/α) ₈ TIM-barrel, PEP/pyruvate superfamily	PMID: 12906829
Quaternary structure	Homodecamer (pentamer of dimers); C-terminal helix swap	PMID: 12842039 ; PMID: 6463
Localization	Cytoplasm	UniProt
Monomer size	266 aa, ~29 kDa	UniProt / sequence
Gene context	panB–panC cluster (PP_4699–PP_4700); folK neighbor	KEGG ppu
Model confidence	AlphaFold pLDDT 97.5 (97% > 90)	AF-Q88DW9-F1 v6

Biological significance

Coenzyme A is an essential, universal cofactor for acyl-group transfer, fatty-acid metabolism, and central carbon metabolism (TCA cycle). Because *P. putida* KT2440 encodes the complete pantothenate → CoA route and PanB is the committed entry point, PanB is expected to be metabolically essential (or conditionally essential in pantothenate-free media). Its position at the valine/branched-chain branch point also means PanB competes for α-ketoisovalerate flux with valine biosynthesis, giving it a regulatory role in balancing amino-acid versus cofactor demands. The absence of a canonical *panD* in the KEGG orthology set suggests *P. putida* obtains β-alanine (needed downstream by PanC) via an alternative route — a downstream detail that does not affect PanB's own function but is worth noting for full pathway reconstruction.

Evidence Base

PMID	Title (abbrev.)	Contribution
12837791	<i>Comparative analysis of the E. coli KPHMT crystal structure ... ($\beta\alpha$)₈ PEP/pyruvate superfamily</i>	Defines the KPHMT reaction (hydroxymethyl transfer onto α -ketoisovalerate) and superfamily membership
6463	<i>Ketopantoate hydroxymethyltransferase II. Physical, catalytic, regulatory properties (Powers & Snell)</i>	Direct kinetics (Km values), Mg^{2+} dependence, Class II aldolase classification, decamer of 10 subunits, pH optimum
12906829	<i>Structure of E. coli KPHMT complexed with ketopantoate and Mg^{2+}</i>	Active-site Mg^{2+} coordination and C3 orientation; ($\beta\alpha$) ₈ fold; 560 kDa decamer
12842039	<i>Crystal structure of KPHMT from M. tuberculosis</i>	First committed step; CoA precursor; decamer via C-terminal helix swapping
18726075	<i>Towards engineering increased pantothenate (vitamin B5) levels in plants</i>	Four-enzyme pathway definition (panB/panD/panC/panE); PanB is first

Supporting/contextual literature reviewed but not directly used for core claims includes reviews on CoA-biosynthesis as an antimicrobial/antiparasitic/antifungal drug target ([PMID: 41967290](#), [PMID: 41914741](#), [PMID: 40801835](#)), which underscore the pathway's essentiality and therapeutic relevance, and a *B. thuringiensis* expression study that identifies a PanB homolog among development-related transcripts ([PMID: 26191639](#)).

Cross-species consistency. The three independently characterized orthologs — *E. coli* (biochemistry + structure), *M. tuberculosis* (structure), and by inference *P. putida* (sequence + AlphaFold) — agree on reaction, cofactor, mechanism, fold, and oligomeric state. No paper challenges the assignment. The main gap is that all direct experimental data derive from orthologs rather than the *P. putida* protein itself.

Limitations and Knowledge Gaps

1. **No direct experimental characterization of the *P. putida* enzyme.** All kinetic, mechanistic, and structural data come from *E. coli* and *M. tuberculosis* orthologs. The *P. putida* assignment rests on strong homology (59.4% identity to *E. coli*), conserved catalytic residues, and a high-confidence AlphaFold model — robust but inferential.
 2. **Substrate-specificity numbers are borrowed.** The reported K_m values (0.18/1.1/5.9 mM for THF/ α -ketoisovalerate/formaldehyde) are *E. coli* values ([PMID: 6463](#)); *P. putida*-specific kinetics have not been measured here.
 3. **β -alanine supply route unresolved.** A canonical *panD* (aspartate-1-decarboxylase) was not retrieved by KEGG orthology in *ppu*, leaving the source of β -alanine (needed by PanC downstream) unclear. This is a pathway-level gap, not a PanB-level one.
 4. **panE assignment is ambiguous.** Three candidate ketopantoate reductase loci (PP_1351/PP_2325/PP_2998) were flagged; the physiologically relevant one is not pinned down.
 5. **Essentiality not experimentally confirmed** in *P. putida* KT2440; it is inferred from pathway completeness and the essential nature of CoA.
 6. **The AlphaFold model is a monomer.** It does not directly validate the decameric assembly or the domain-swapped inter-subunit interface, which are inferred from orthologous crystal structures.
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Proposed Follow-up Experiments / Actions

1. **Recombinant expression and enzymatic assay.** Clone PP_4699, purify the His-tagged protein, and measure KPHMT activity spectrophotometrically (or by coupled assay), determining *P. putida*-specific K_m/k_{cat} for methylene-THF and α -ketoisovalerate and confirming the Mg^{2+} requirement.
2. **Confirm quaternary structure.** Use size-exclusion chromatography with multi-angle light scattering (SEC-MALS) or native mass spectrometry to verify the ~290 kDa homodecamer, and consider cryo-EM/crystallography to confirm the domain-swapped assembly.
3. **Site-directed mutagenesis** of Asp45, Asp84, Glu114, and Glu181 to validate their predicted roles in Mg^{2+} /substrate binding and general-base catalysis.

4. **Gene-essentiality test.** Attempt markerless deletion of PP_4699 and test for pantothenate auxotrophy (rescue by exogenous pantothenate or β -alanine) to confirm the committed-step, essential role.
 5. **Resolve the β -alanine and panE gaps.** Use transcriptomics/metabolomics or targeted knockouts to identify the β -alanine source and the operative ketopantoate reductase among PP_1351/PP_2325/PP_2998.
 6. **Structural biology on the native protein.** Determine an experimental structure of the *P. putida* enzyme, ideally with bound Mg^{2+} and ketopantoate, to directly validate the AlphaFold active-site model.
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Report compiled from 5 investigation iterations, 6 confirmed findings, and 12 reviewed papers. Gene identity verified against UniProt Q88DW9 and 12 concordant signature databases; no conflicting family assignment was found.
